Lakshya JEE 2.0 (2024)

Definite Integration

DPP-01

1. If $\int_0^p \frac{dx}{1+4x^2} = \frac{\pi}{8}$, then the value of p is

(1) $\frac{1}{4}$ (2) $-\frac{1}{2}$

(3) $\frac{3}{2}$ (4) $\frac{1}{2}$

2. $\int_0^1 \frac{1}{3+4x}\,dx =$

(1) $ln\left(\frac{7}{3}\right)$ (2) $4ln\left(\frac{7}{3}\right)$

(3) $\frac{1}{4}ln(7/3)$ (4) $ln\,(3/7)$

3. The value of $\int_0^1 \frac{x^4+1}{x^2+1}\,dx$ is

(1) $\frac{1}{6}(3-4\pi)$ (2) $\frac{1}{6}(3\pi+4)$

(3) $\frac{1}{6}(3+4\pi)$ (4) $\frac{1}{6}(3\pi-4)$

4. $\int_0^4 \frac{dx}{\sqrt{8x-x^2}}\,dx$

(1) 0 (2) $\frac{\pi}{4}$

(3) $\frac{\pi}{2}$ (4) π

5. $\int_0^{\ln 2} x\,e^{-x}dx =$

(1) $\frac{-1}{2}\ln(2e)$ (2) $\frac{1}{2}\ln\left(\frac{e}{2}\right)$

(3) $\frac{1}{2}\ln(2/e)$ (4) $\frac{1}{2}\ln(2e)$

6. If $f(x) = \tan x - \tan^3 x + \tan^5 x - \ldots\infty$ with $0 < x < \frac{\pi}{4}$, then $\int_0^{\pi/4} f(x)dx$ is equal to:

(1) 1 (2) 0

(3) $\frac{1}{4}$ (4) $\frac{1}{2}$

7. $\int_0^1 x^2 \cos x^3 dx$

(1) 3 sin1 (2) $\frac{\sin 1}{3}$

(3) $\sin\left(\frac{1}{3}\right)$ (4) None of these

8. $\int_0^{\pi/4} \frac{\sin 2x\,dx}{\sin^4 x + \cos^4 x}\,dx$

(1) $\frac{\pi}{4}$ (2) $\frac{\pi}{2}$

(3) π (4) 2π

9. $\int_0^{\pi/2} \frac{\sin x - \cos x}{(1+\sin x\cos x)}\,dx$

(1) 0 (2) 1

(3) –1 (4) None of these

10. $\int_0^1 \frac{\sin^{-1}\sqrt{x}}{\sqrt{x(1-x)}}\,dx$

(1) $\frac{\pi}{4}$ (2) $\frac{\pi}{2}$

(3) $\frac{\pi^2}{4}$ (4) $\frac{\pi^2}{16}$

11. If $f(x)$ is monotonic differentiable function on $[a, b]$, then $\int_a^b f(x)dx + \int_{f(a)}^{f(b)} f^{-1}(x)dx =$

(1) $bf(a) - af(b)$

(2) $bf(b) - af(a)$

(3) $f(a) + f(b)$

(4) Cannot be found

12. Let $\frac{d}{dx}F(x) = \left(\frac{e^{\sin x}}{x}\right); x > 0$.

If $\int_1^4 \frac{3}{x} e^{\sin x^3} dx = F(k) - F(1)$, then one of the possible value of k, is

(1) 15 (2) 16

(3) 63 (4) 64

13. The value of $\int_1^2 \frac{dx}{x(1+x^4)}$ is equal to

(1) $\frac{1}{4}\log\frac{17}{32}$ (2) $\frac{1}{4}\log\frac{17}{2}$

(3) $\log\frac{17}{2}$ (4) $\frac{1}{4}\log\frac{32}{17}$

14. If $f(x) = x^3 + 3x + 4$ then the value of $\int_{-1}^{1} f(x)dx + \int_0^4 f^{-1}(x)dx$ equals:

(1) 4 (2) $\frac{17}{4}$

(3) $\frac{21}{4}$ (4) $\frac{23}{4}$

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (4)
2. (3)
3. (4)
4. (3)
5. (2)
6. (3)
7. (2)
8. (1)
9. (1)
10. (3)
11. (2)
12. (4)
13. (4)
14. (4)

PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if